

Farm Assist

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Agriculture is one of the most important sectors of the economy, yet farmers often face difficulties in accessing accurate and timely information related to crop cultivation, plant diseases, fertilizer recommendations, and modern farming practices. Existing agricultural information is scattered across multiple sources such as websites, books, and advisory platforms, making it time-consuming and challenging for farmers to find reliable answers when they need them. To address this issue, **Farm Assist** is proposed as a centralized agricultural information system that provides fast, reliable, and easy access to comprehensive crop-related knowledge through a single searchable platform.

The system consolidates essential information, including crop details, disease identification, fertilizer guidance, pest management, cultivation techniques, and best farming practices into a well-organized knowledge base. By leveraging efficient **Data Structures and Algorithms (DSA)** for data organization, searching, and retrieval, Farm Assist delivers relevant information quickly and accurately, even when handling large datasets. This improves the speed and efficiency of information access while ensuring a user-friendly experience.

The primary objective of Farm Assist is to support farmers in making informed decisions, increasing agricultural productivity, reducing crop losses caused by diseases, and promoting the adoption of modern and sustainable farming techniques. By providing reliable agricultural knowledge at the right time, the system serves as a valuable digital assistant that enhances farming efficiency, encourages better resource utilization, and contributes to the overall growth of the agricultural sector.

Signature of the Guide